Introduction to R

Advanced Statistics Lab

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Theme

Get R

From CRAN

RStudio

Ubuntu:

```
sudo apt install r-base
```

Some Introduction Material

From the developers: Introduction to R

RStudio: Documentation

R for Data Science

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Background

- R is a general programming language with a specialization for statistics.
- R is generally used through a REPL and interpreted live.
- Functional programming paradigm: all about objects and functions.
- Major appeal: packages, graphics, and scientific workflows.
- Major drawback: scripts can grow without structure.

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Assigning And Manipulating Variables

Note: R has two assignment operators, `<-` and `=`.

```r
x = 5
print(x)
```

```
[1] 5
```

```r
x
```

```
[1] 5
```

```r
y = x
print(y)
```

```
[1] 5
```

```r
x = 3

print(x)
```

```
[1] 3
```

```r
print(y)
```

```
[1] 5
```

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Working With Data Frames

```r
dat = data.frame(
  name = c("Adam", "Paul", "Elly"),
  age = c(22, 43, 54)
)

str(dat)
```

```
'data.frame':	3 obs. of  2 variables:
 $ name: chr  "Adam" "Paul" "Elly"
 $ age : num  22 43 54
```

```r
names = dat$name
ages = dat[, "age"]

str(names)
```

```
 chr [1:3] "Adam" "Paul" "Elly"
```

```r
str(ages)
```

```
 num [1:3] 22 43 54
```

Subsetting Data Frames

```r
dat[dat$name == "Paul", ]
```

```
  name age
2 Paul  43
```

```r
dat[dat$age %in% c(22, 54), ]
```

```
  name age
1 Adam  22
3 Elly  54
```

```r
dat$id = 1:nrow(dat)
dat[, "id_2"] = dat$id + 2

print(dat)
```

```
  name age id id_2
1 Adam  22  1    3
2 Paul  43  2    4
3 Elly  54  3    5
```

```r
dat$id[dat$name == "Adam"] <- 12
print(dat)
```

```
  name age id id_2
1 Adam  22 12    3
2 Paul  43  2    4
3 Elly  54  3    5
```

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Reading And Writing Data

```r
dat = data.frame(
  name = c("Adam", "Paul", "Elly"),
  age = c(22, 43, 54)
)

write.table(dat, file = "my_data.csv", row.names = FALSE, sep = ",")

dat_read = read.table(file = "my_data.csv", sep = ",", header = TRUE)
str(dat_read)
```

```
'data.frame':	3 obs. of  2 variables:
 $ name: chr  "Adam" "Paul" "Elly"
 $ age : int  22 43 54
```

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Vectors And Matrices

```
x = as.integer(1)
str(x)
```

```
 int 1
```

```
is.vector(x)
```

```
[1] TRUE
```

```
y = rnorm(10)
str(y)
```

```
 num [1:10] -0.775 -0.503 1.123 -0.824 1.583 ...
```

```
D = matrix(data = rnorm(5 * 5), nrow = 5, ncol = 5)
D
```

```
           [,1]       [,2]        [,3]       [,4]       [,5]
[1,] -1.0947200 -1.6864584 -0.36647287 -0.4245857 -0.1828272
[2,] -1.0048981 -0.2281715 -0.04786797  0.7488563 -0.6177607
[3,] -0.6136573  0.6747762 -0.29883223 -0.4020819  0.4380955
[4,]  0.4388820 -1.2880571  0.20272960 -0.6785895  0.4287005
[5,]  0.2703989 -0.4033161 -0.19445280  0.1717627  1.3157590
```

```
class(D)
```

```
[1] "matrix" "array"
```

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Matrix Helpers

```
means = colMeans(D)
means
```

```
[1] -0.4007989 -0.5862454 -0.1409793 -0.1169276  0.2763934
```

```
d = diag(D)
d
```

```
[1] -1.0947200 -0.2281715 -0.2988322 -0.6785895  1.3157590
```

Data Exploration And Visualization

```
library(dlookr)
library(ggplot2)

data(iris)
str(iris)
```

```
'data.frame':	150 obs. of  5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species     : Factor w/ 3 levels "setosa","versicolor",..: 1 1 1 1 1 1 1 1
1 1 ...
```

```
dlookr::diagnose(iris)
```

```
# A tibble: 5 × 6
  variables    types   missing_count missing_percent unique_count unique_rate
  <chr>        <chr>           <int>           <dbl>        <int>       <dbl>
1 Sepal.Length numeric             0               0           35       0.233
2 Sepal.Width  numeric             0               0           23       0.153
3 Petal.Length numeric             0               0           43       0.287
```

```
4 Petal.Width  numeric          0          0      22     0.147
5 Species      factor           0          0       3     0.02
```

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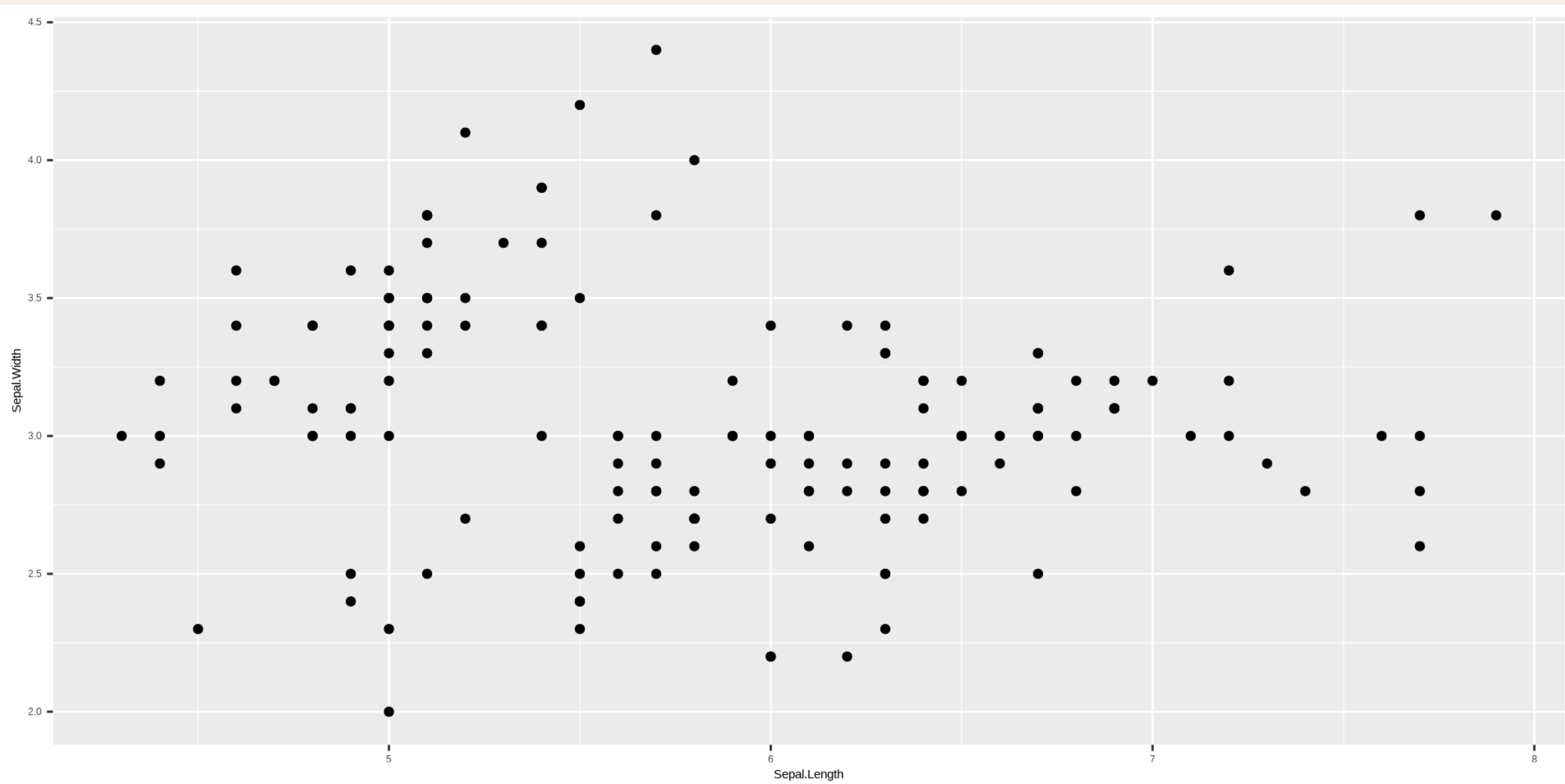
Scatterplot

```
p = ggplot(iris, aes(x = Sepal.Length, y = Sepal.Width)) +
  geom_point()

p
```



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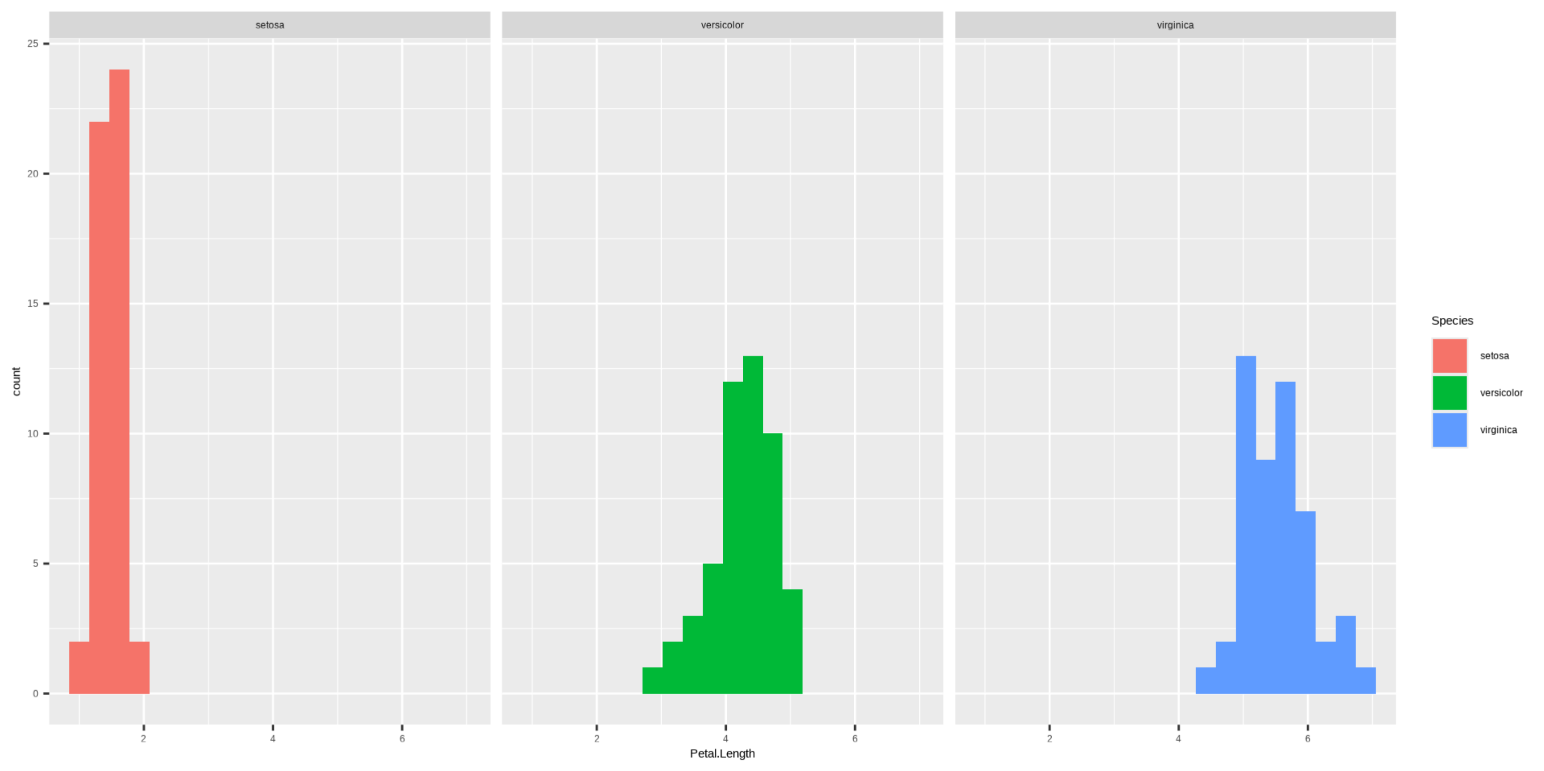
Histogram

```
p2 = ggplot(iris, aes(x = Petal.Length)) +
  geom_histogram(aes(fill = Species), bins = 20) +
  facet_wrap(~ Species)

p2
```



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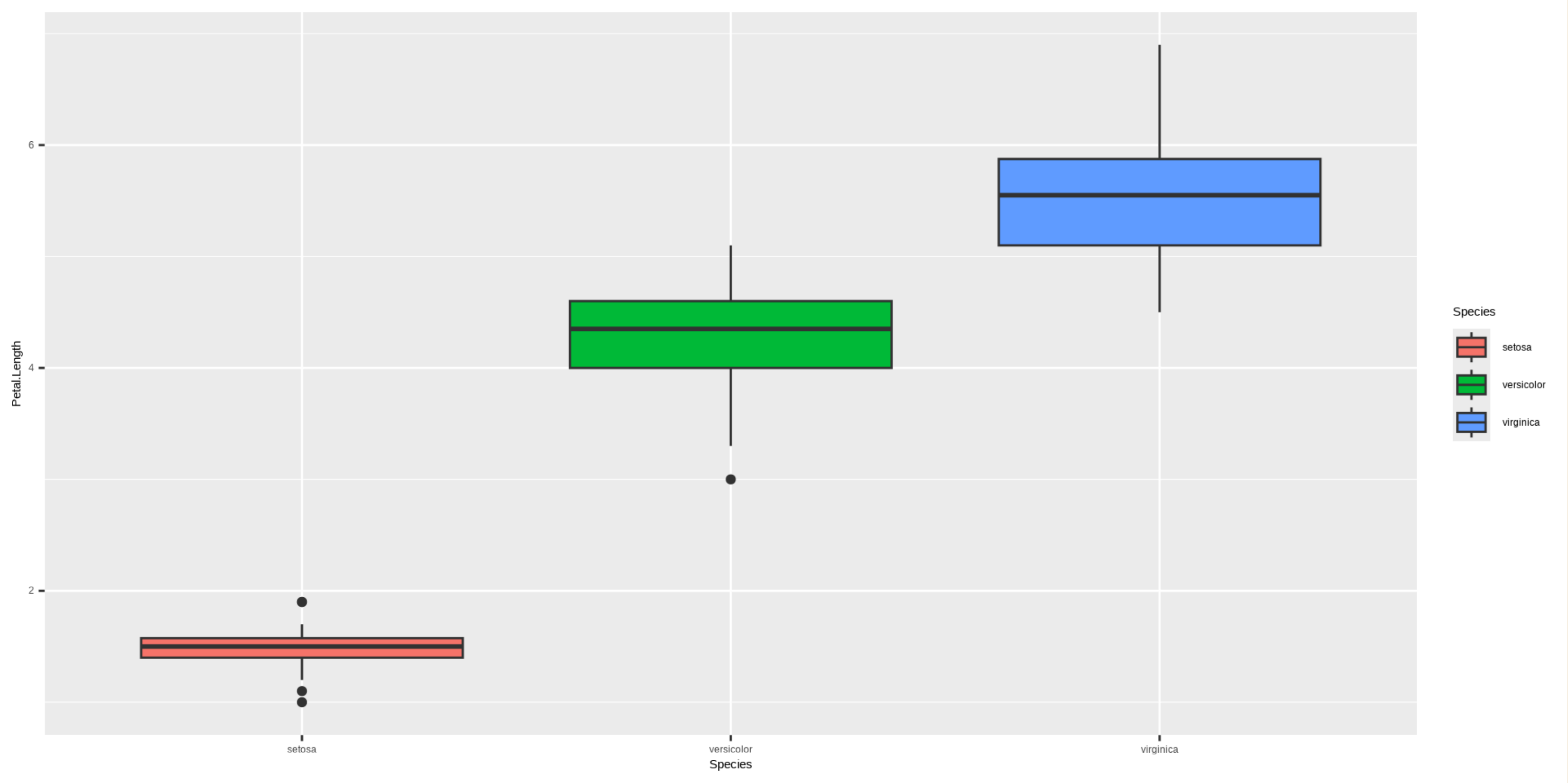
Boxplot

```
p3 = ggplot(iris, aes(x = Species, y = Petal.Length, fill = Species)) +
  geom_boxplot()

p3
```



Writing Functions

All operators in R are functions.

```
?sum
?`+`
fix(`+`)
```

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A Mean Function

```r
my_mean_function = function(y) {

  n = length(y)
  sum = 0
  for (i in 1:n) sum = sum + y[i]

  mean = sum / n

  return(mean)

}
```

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Use The Function

```r
x = rnorm(1000, mean = 23.42, sd = 4)

our_mean = my_mean_function(x)
our_mean
```

```
[1] 23.31251
```

```r
mean(x)
```

```
[1] 23.31251
```